

$$\ddot{\theta}I + \dot{\theta}b + \theta k = \tau$$



Anthropometric information

Sex

Age

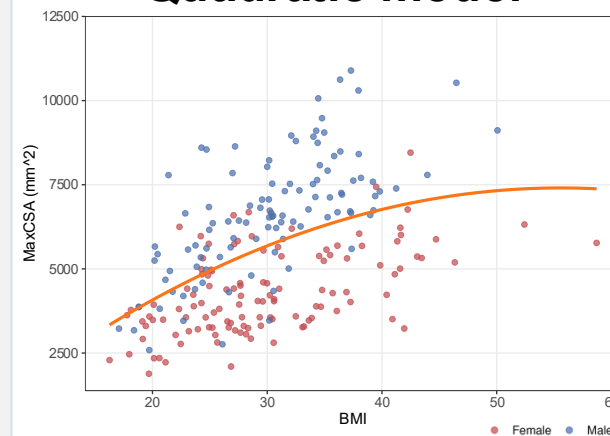
BMI

Mass

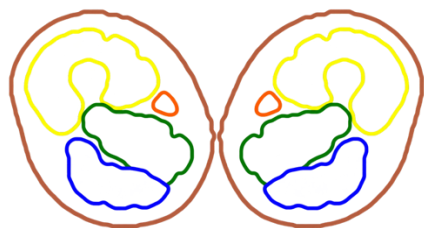
Height

Others

Quadratic model

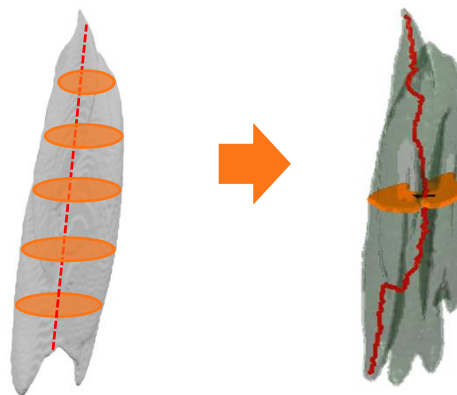


CT data



Segmentation by MATLAB pipeline

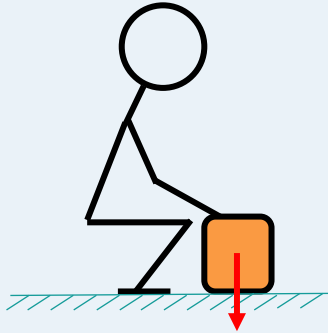
Muscle
STLs



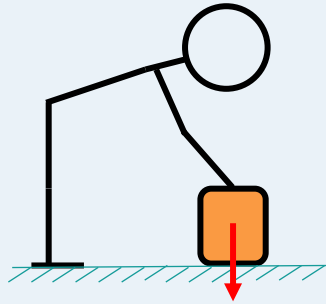
Max Slice



Squat Lift



Stoop Lift



Biomechanical Modeling

3DSSPP

Static
posture-based

L4/L5 compression

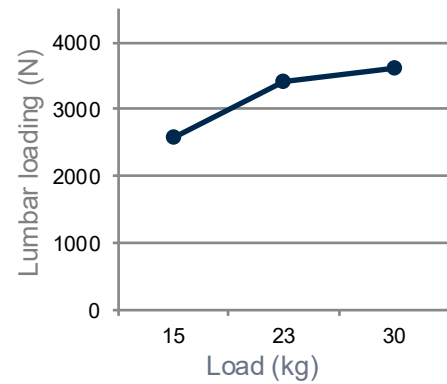
OpenSim

Dynamic
time-based

Joint reaction force

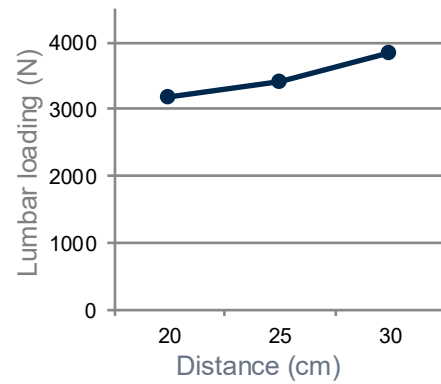
Task Parameters

Hand Load



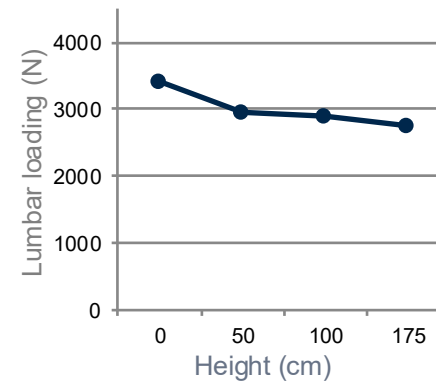
Heavier hand load
→ higher lumbar loading

Horizontal Distance



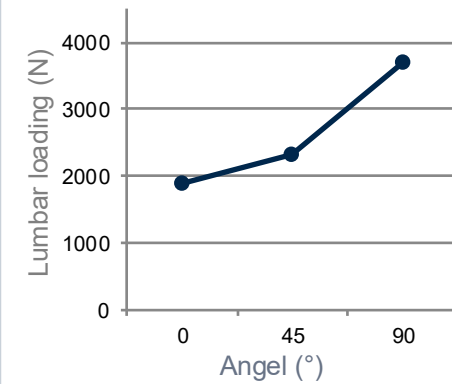
Farther from body
→ higher lumbar loading

Vertical Height



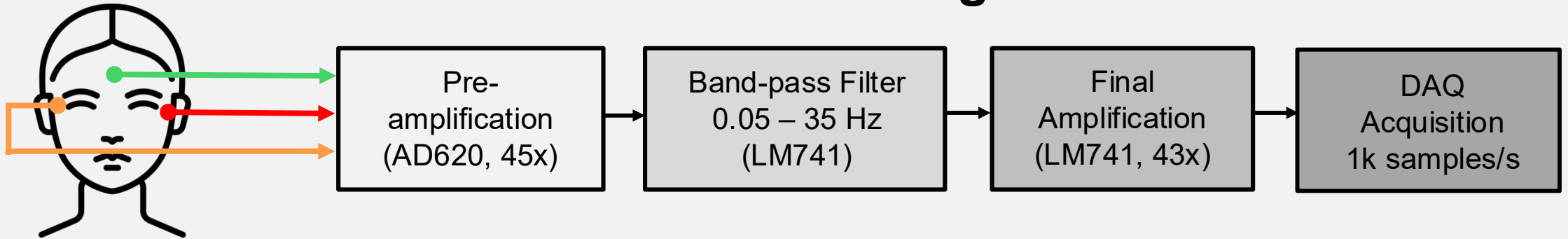
Higher position
→ lower lumbar loading

Asymmetry Angle



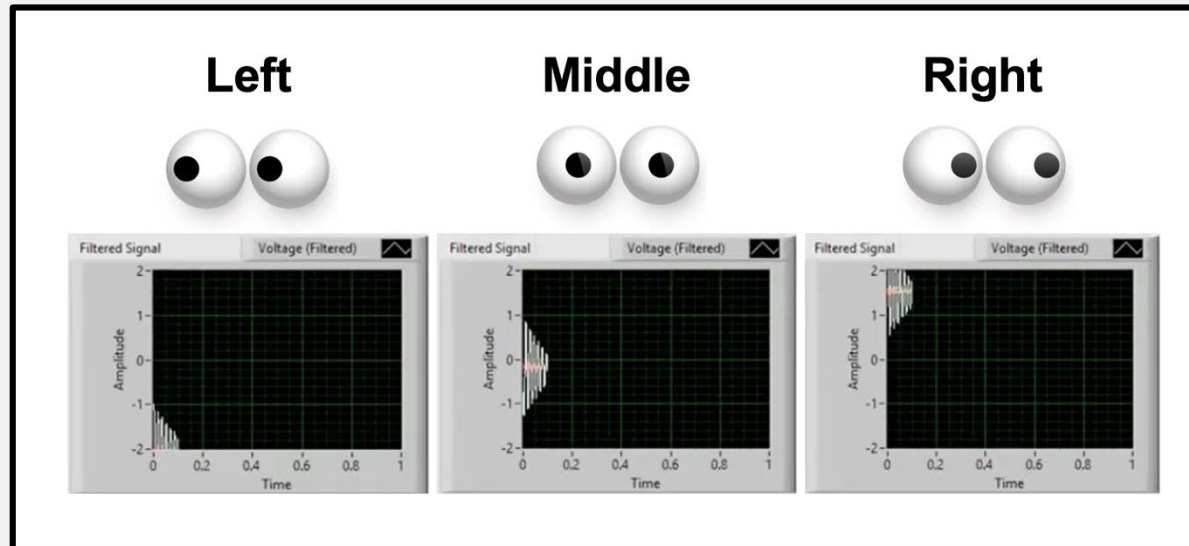
Twisting
→ higher lumbar loading

Conditioning

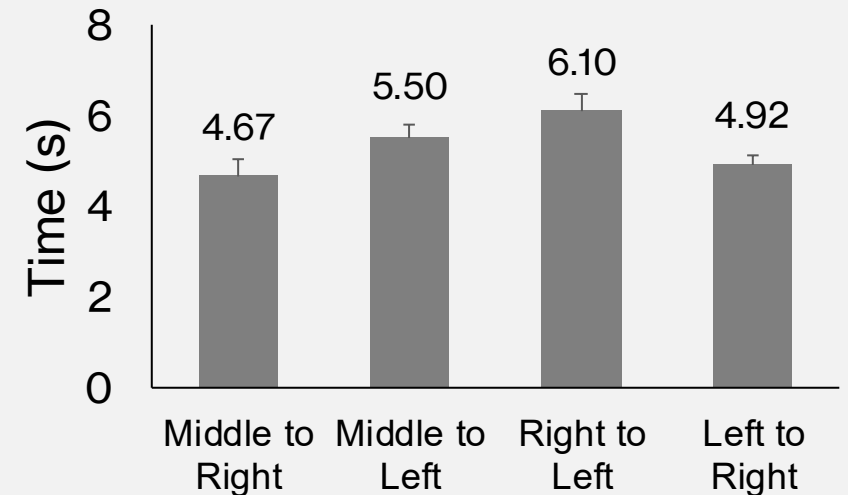


Acquisition and Results

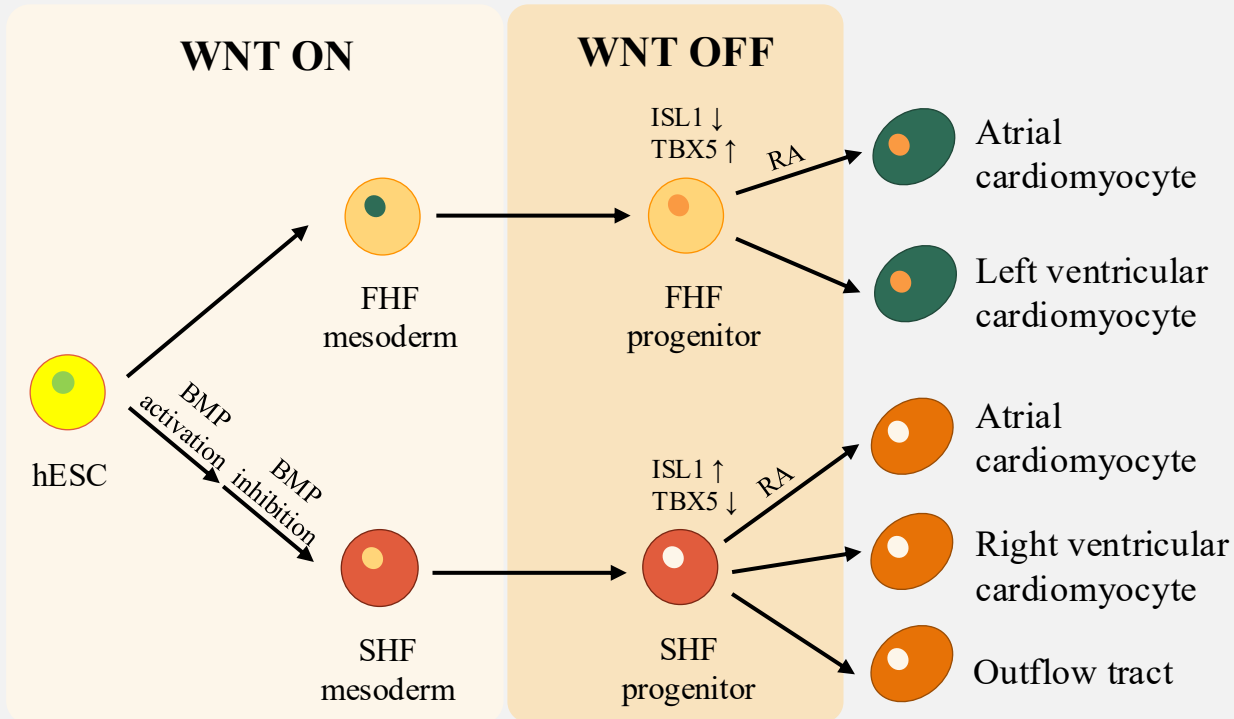
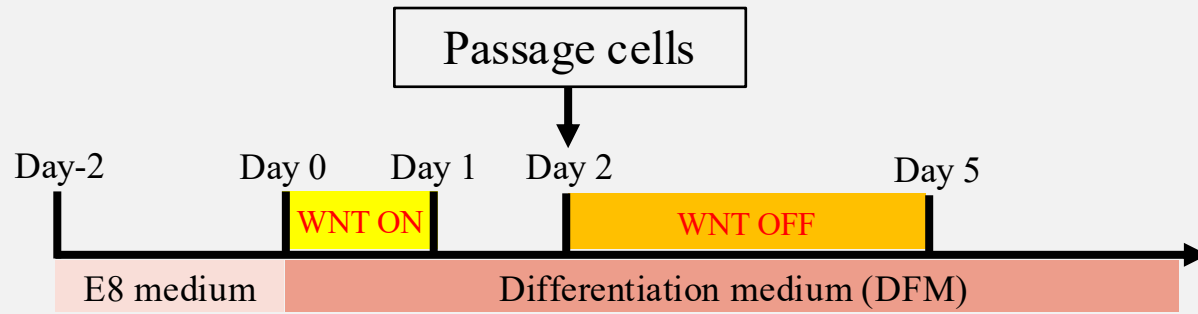
Recognize Eye Movement by LabVIEW



System Response Time by Eye Movement

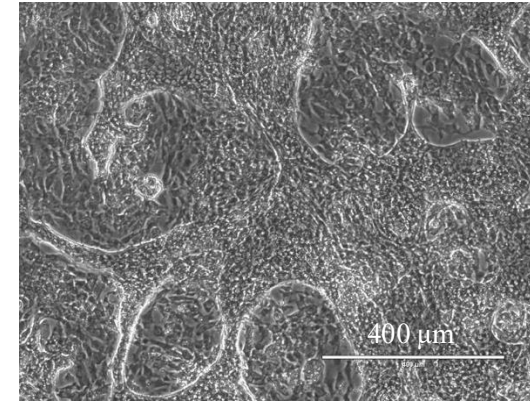


Culture Timeline

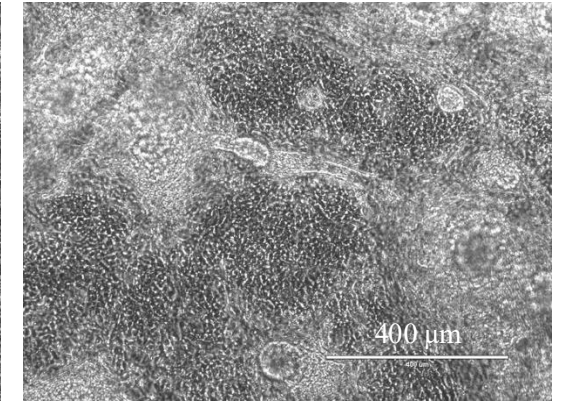


Characterization

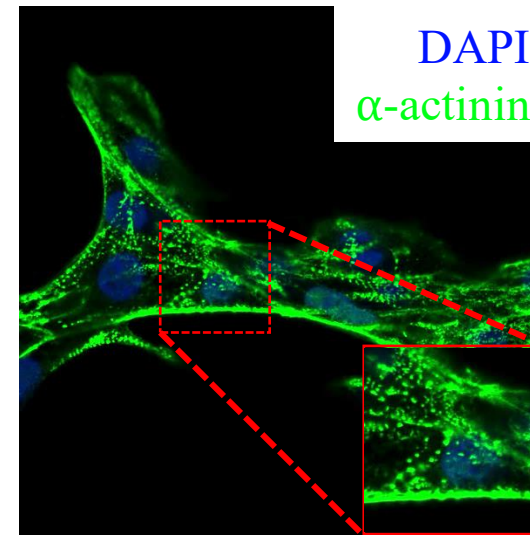
No passaged cell



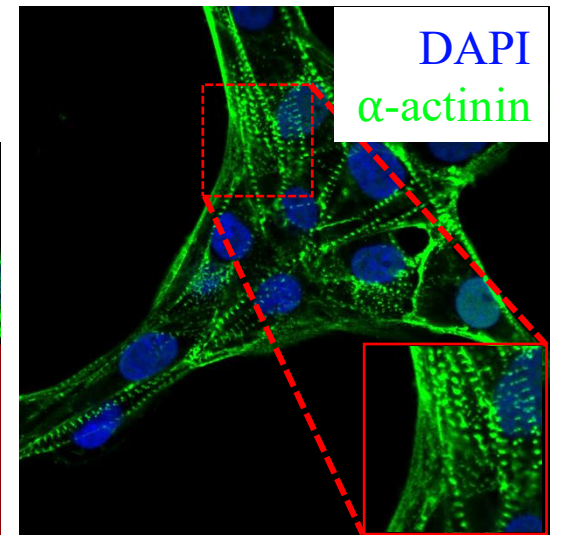
Passaged cell

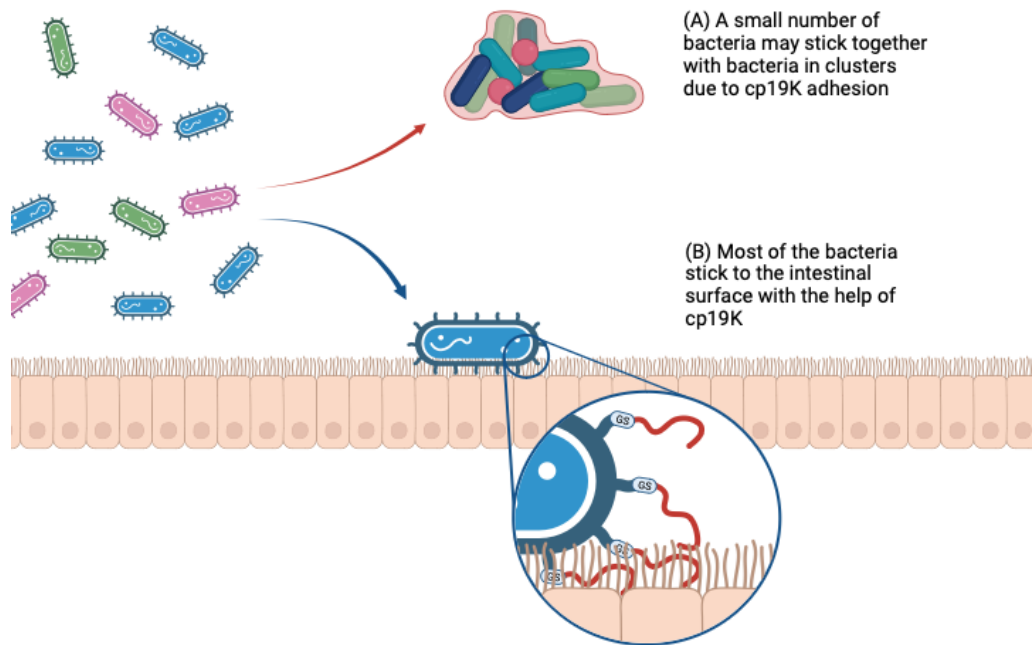


Cardiomyocyte

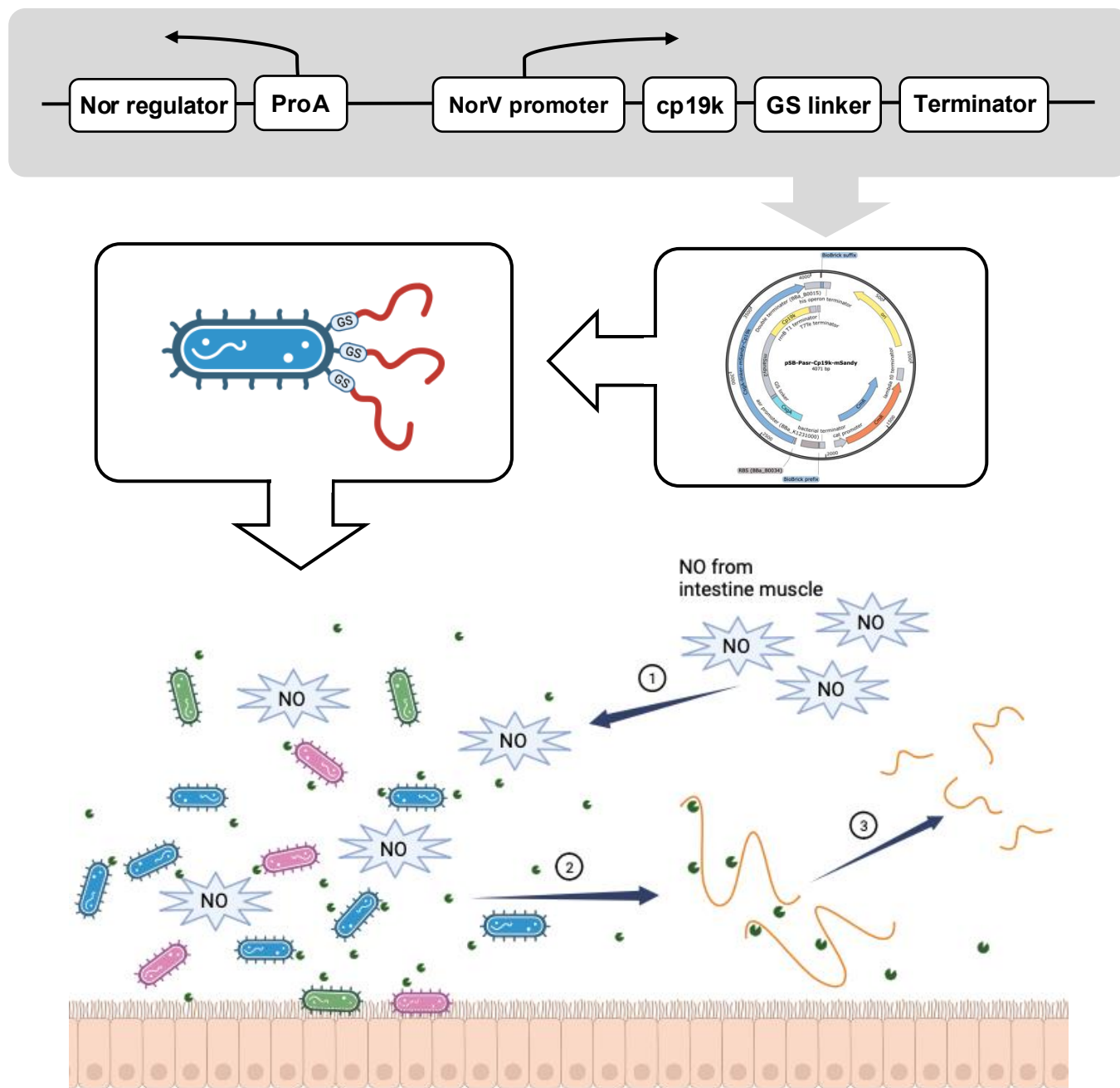
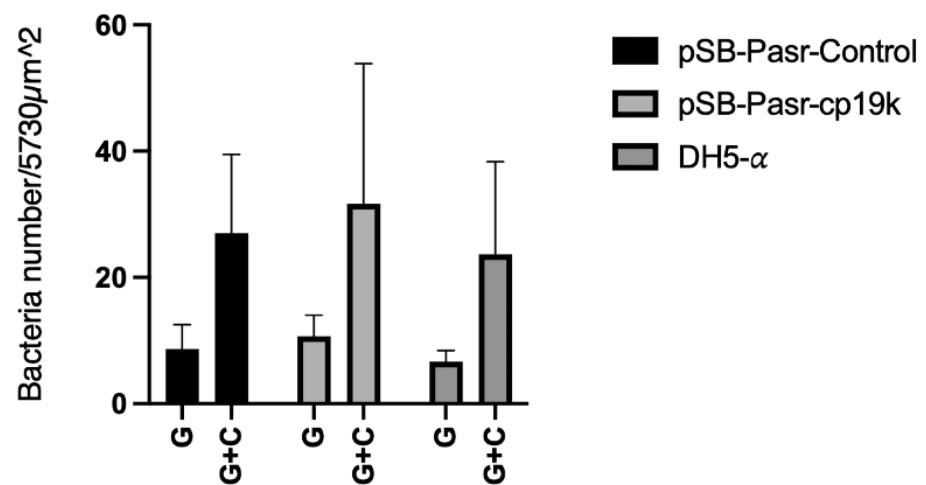


Atrial cardiomyocyte



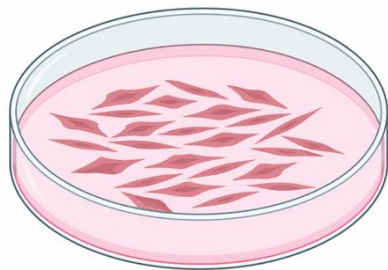


Bacteria adhere number after 1 h of culture (pH=5)



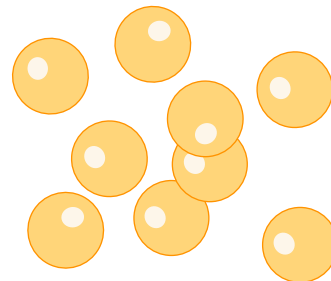
In vitro

MyoD^{-/-} C2C12 myoblast

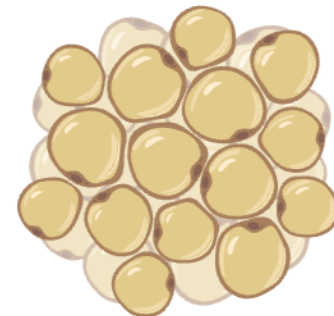


Insulin + Dex + IBMX
+ rosiglitazone

PRDM16 / CEBP β



PPAR γ / UCP1



In vivo

6-week-old C57BL

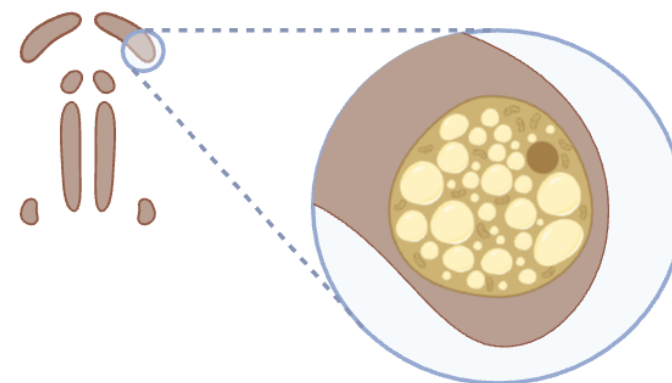


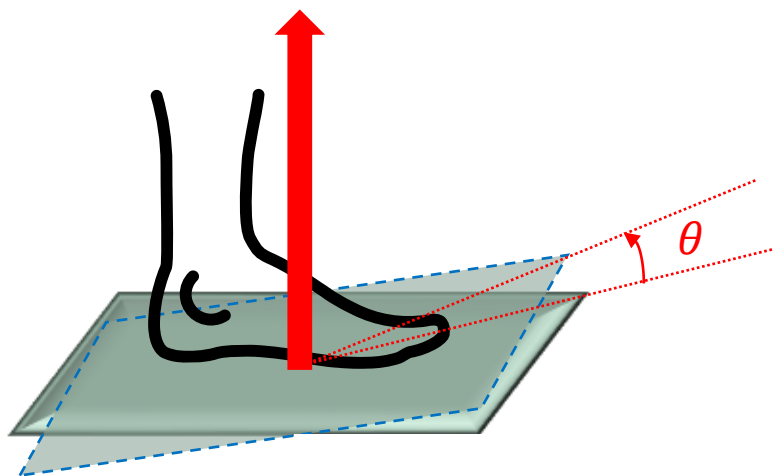
High-fat diet

T2D mice



Brown adipose tissue





$$\ddot{\theta}I + \dot{\theta}b + \theta k = \tau$$

